

SPECIAL EVENTS

PHASES OF THE MOON

	Full Moon	New Moon
2008	June 18	June 3
2009	June 7	June 22
2010	June 26	June 12
2011	June 15	June 1
2012	June 4	June 19

ECLIPSES

2010: June 26 A partial eclipse of the Moon is visible from the Pacific Ocean, Australasia, and eastern Asia.

2011: June 1 A partial eclipse of the Sun is visible from Greenland, northern North America, and parts of northern and northeastern Asia.

2011: June 15 A total eclipse of the Moon is visible from Australia, southern Asia, the Indian Ocean, Africa, and Europe.

2012: June 4 A partial eclipse of the Moon is visible from western North and South America, the Pacific Ocean, Australasia, and eastern Asia.

THE PLANETS

2009: June 13 Mercury is at greatest morning elongation, magnitude 0.6.

2009: June 21–22 Venus and Mars appear four Moon-widths apart in the eastern dawn sky.

2012: June 5–6 A transit of Venus across the Sun is visible from North America, the Pacific Ocean, Australasia, and Asia.

2012: June 30 Venus and Jupiter appear about ten Moon-widths apart in the eastern dawn sky.

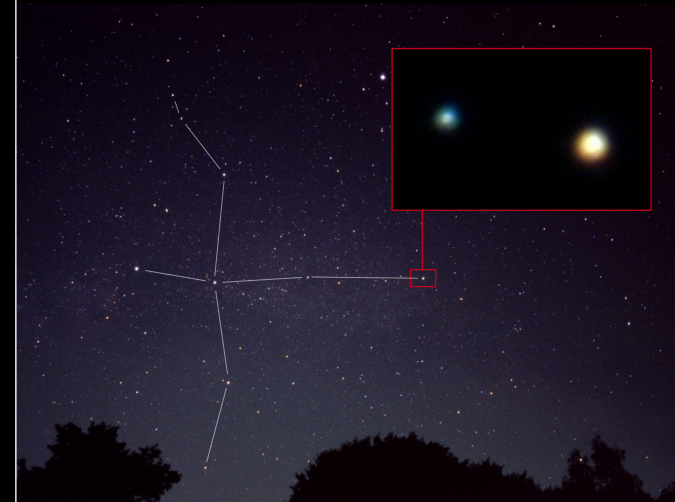
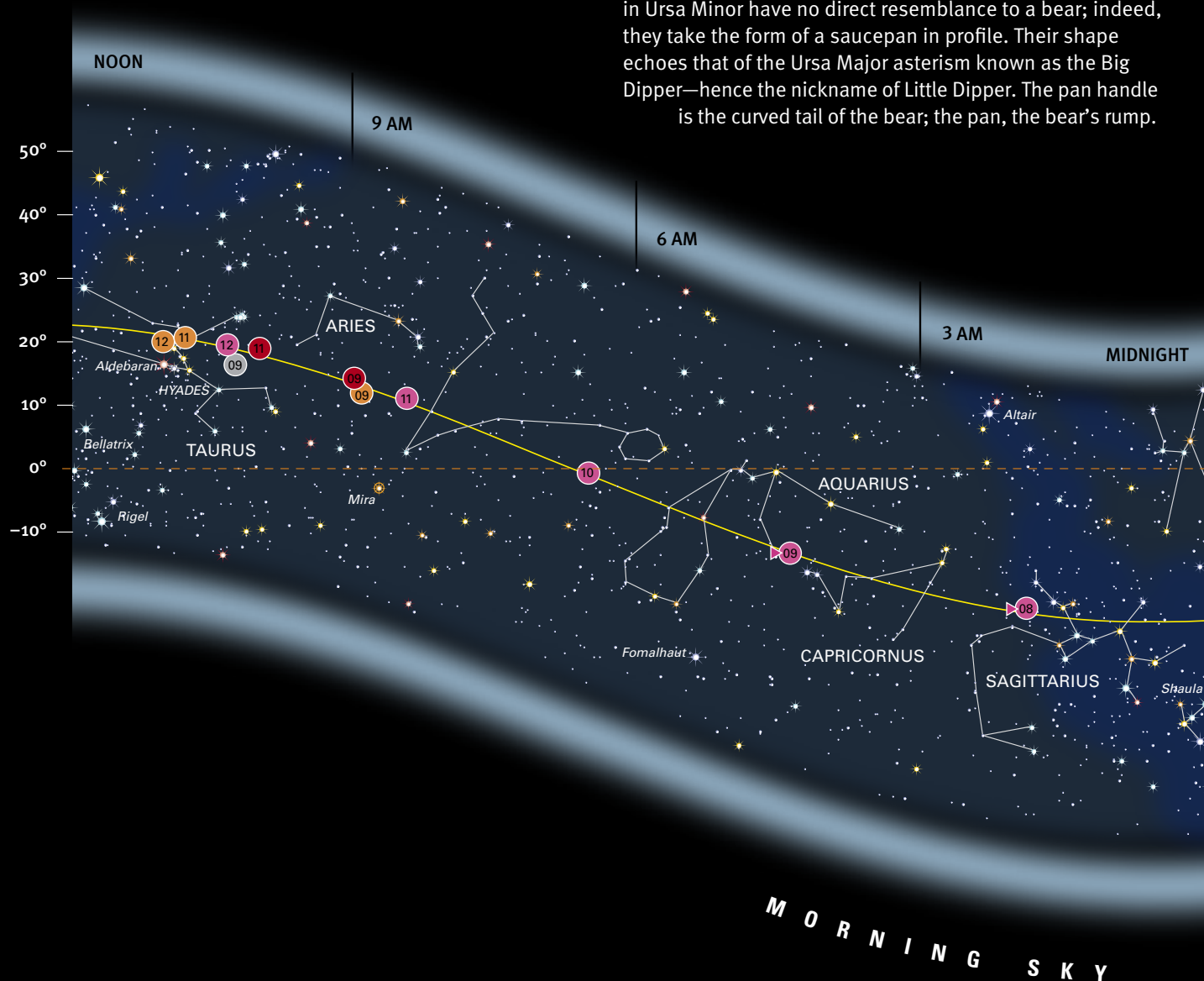
Elongation is explained on *page 14*

June

The mythical hero Hercules moves into the June sky to join Boötes, the herdsman, and Ophiuchus, the serpent-holder. Toward the end of the month, the nights are at their shortest for northern observers and at their longest for those in the southern hemisphere. This is because on June 21, the Sun is at its farthest point north of the celestial equator.

NORTHERN LATITUDES

When looking south, Hercules and Boötes are high in the sky. The bright star Arcturus makes Boötes easy to find. Hercules, although large, is not prominent. It is best found by locating Vega in the east. Vega is in the constellation Lyra and is the fifth-brightest star of all. It is outshone in the June sky only by Arcturus. To the right of Vega is the body of Hercules. Four linked stars that form a distorted square, called the Keystone, represent his lower torso.



Albireo
The second-brightest star in Cygnus, Albireo marks the swan's beak. It appears as a single star with the naked eye, but when viewed with powerful binoculars or a telescope, it is seen as a double star.

His legs point upward, and his head points toward the horizon. The star cluster M13 can be spotted with the naked eye and lies on a line linking two of the stars. The head of Scorpius is just above the horizon, located by spotting red supergiant Antares.

In the east are three stars that form the asterism known as the Summer Triangle. The brightest and highest is Vega, and nearer to the horizon, Deneb (Cygnus) and Altair (Aquila) can be seen. Due north, Ursa Minor extends above Polaris, the Pole Star, which marks the tip of the bear's tail. The stars in Ursa Minor have no direct resemblance to a bear; indeed, they take the form of a saucepan in profile. Their shape echoes that of the Ursa Major asterism known as the Big Dipper—hence the nickname of Little Dipper. The pan handle is the curved tail of the bear; the pan, the bear's rump.